Statistics for International Commerce

Week 9: One-Sample Tests of Hypothesis

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Hello everyone!

Yet another **great day** to *keep learning* statistics for international commerce. :-)

Agenda

Hypothesis Testing

- From estimation → to **decision making**
- One-sample tests
- z-test vs t-test
- p-value
- Type I and Type II errors

Learning Objectives

By the end:

- understand **null vs alternative hypothesis**
- perform **one-sample tests**
- interpret **p-values**
- make **business decisions using data**

Motivation (Business Context)

A company claims:

> "Average delivery time is under 5 days."

We have sample data.

Question:

Do we believe the claim?

What is a Hypothesis?

A hypothesis is a **testable statement** about a population.

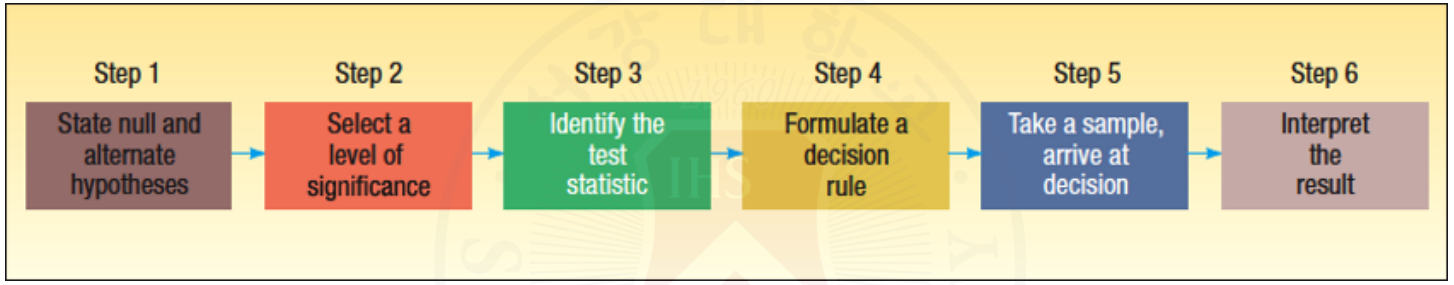
Examples:

- Mean delivery time = 5 days
- Average spending = 60 USD
- Return rate = 70%

Hypothesis Testing

Definition:

A procedure using **sample data + probability** to decide *whether a claim is reasonable*.



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Step 1: Define Hypotheses

Two competing claims about a population parameter.

	Hypothesis	Role
H_0	Null	The "nothing is happening" claim
H_1	Alternative	What we want to find evidence for

> We **never prove** H_0 — we either reject it or fail to reject it.

Null Hypothesis H_0

- The **default** assumption — assume nothing has changed
- Contains an **equality** (=, ≤, ≥)
- We assume it is true until the data say otherwise

Commerce examples:

- $H_0 : \mu = 5$ days (delivery time is as claimed)
- $H_0 : \mu = 100$ USD (average order has not changed)
- $H_0 : \mu \leq 50$ kg (container weight is within limit)

Alternative Hypothesis H_1

- The claim we are **trying to support with data**
- Contains a strict inequality (≠, <, >)
- Drives which tail of the distribution we look at

Commerce examples matching the H_0 above:

- $H_1 : \mu \neq 5 \rightarrow$ delivery time is **different** (two-sided)
- $H_1 : \mu > 100 \rightarrow$ orders are **larger than claimed** (right-tailed)
- $H_1 : \mu > 50 \rightarrow$ containers are **overweight** (right-tailed)

How to Choose the Alternative?

Ask: **"What would concern us?"**

Situation	H_1	Test
Any difference matters	$\mu \neq \mu_0$	Two-sided
We fear it's too low	$\mu < \mu_0$	Left-tailed
We fear it's too high	$\mu > \mu_0$	Right-tailed

Rule: The alternative reflects the **research question**, not the data.

Key Definitions

NULL HYPOTHESIS (H_0): A statement about the value of a population parameter developed for the purpose of testing numerical evidence.

ALTERNATIVE HYPOTHESIS (H_1): A statement that is accepted if the sample data provide sufficient evidence that the null hypothesis is false.

> H_0 and H_1 are **mutually exclusive** and **collectively exhaustive**.

Example: $H_0 : \mu = 0$ vs. $H_1 : \mu \neq 0$

mutually exclusive and **collectively exhaustive** means that exactly one of the hypotheses must be true. They cannot both be true at the same time, and they cover all possible outcomes. In this example, either `(\mu)` is equal to 0 (H_0 is true) or `(\mu)` is not equal to 0 (H_1 is true). There are no other possibilities.

Hypothesis

Examples

- *Null Hypothesis*: H_0: There is no difference in the salary of factory workers based on gender.
- *Alternative Hypothesis*: H_1: Male factory workers have a higher salary than female factory workers.
- *Null Hypothesis*: H_0: There is no relationship between height and shoe size.
- *Alternative Hypothesis*: H_1: There is a positive relationship between height and shoe size.
- *Null Hypothesis*: H_0: Experience on the job has no impact on the quality of a brick mason's work.
- *Alternative Hypothesis*: H_1: The quality of a brick mason's work is influenced by on-the-job experience.

Key Reminders: H_0 and H_1

- H_0 is the null hypothesis; H_1 is the alternative hypothesis.
- They are **mutually exclusive** and **collectively exhaustive**.
- We start by assuming H_0 is true; data are used to test this assumption.
- Decisions are based on a random sample of size n.
- **Fail to reject H_0** does not mean H_0 is true; it means evidence is not strong enough to reject it.
- **Reject H_0** means the data support H_1, with Type I error risk controlled by α.
- Equality belongs in H_0 ($=$, $\leq$, $\geq$); strict inequality belongs in H_1 ($\neq$, $<$, $>$).

Types of Tests

Type	Form	When to use	Commerce example
Two-sided	$\mu \neq \mu_0$	Any change matters (up **or** down)	Has avg. order value changed from $100?
Left-tailed	$\mu < \mu_0$	We fear the value is **too low**	Is avg. delivery faster than claimed 5 days?
Right-tailed	$\mu > \mu_0$	We fear the value is **too high**	Are containers heavier than the 50 kg limit?

Step 2: Significance Level α

The **significance level** α is the probability of rejecting H_0 when it is actually true:

$$P(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha$$

Key points:

- Also called the **Type I Error** rate and **false positive** rate
- Must be chosen **before** collecting data — never after!
- Common choices: $\alpha = 0.05$ or $\alpha = 0.01$

> **Intuition:** $\alpha = 0.05$ means we accept a 5% chance of wrongly rejecting a true H_0.

α	Strictness	Typical use
0.10	Lenient	Exploratory research
0.05	**Standard**	**Most business decisions**
0.01	Strict	High-stakes (regulatory, medical)

Another Possible Error: Type II Error (β)

Type II Error is the probability of **not rejecting** H_0 when H_0 is actually false:

$$P(\text{do not reject } H_0 \mid H_0 \text{ is false}) = \beta$$

Key points:

- Also called the **false negative** error
- We do **not** choose β directly
- β depends on α, sample size n, and the true effect size

The **power of a test** is:

$$1 - \beta = P(\text{reject } H_0 \mid H_0 \text{ is false})$$

> **Intuition:** Higher power means a better chance of detecting a real effect.

Errors in Hypothesis Testing

Null Hypothesis	Researcher: Does Not Reject H_0	Researcher: Rejects H_0
H_0 is true	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Type I and Type II Errors

	H_0 is true	H_0 is false
Reject H_0	Type I Error (`(\alpha)`)	Correct decision ✓
Do not reject H_0	Correct decision ✓	Type II Error (`(\beta)`)

Type I Error: Reject H_0 when it is actually true. Probability = α

Type II Error: Fail to reject H_0 when it is actually false. Probability = β

Power of the test = $1 - \beta$ = $P(\text{reject } H_0 \mid H_0 \text{ is false})$

> We choose α. The value of β depends on α, sample size n, and the true parameter value.

Business Interpretation of Errors

Scenario: A logistics firm claims mean delivery time = 5 days. You test this with sample data.

Type I Error (α) — false alarm:

- You conclude delivery times have changed, but they actually have not
- Consequence: unnecessary investment in process overhaul

Type II Error (β) — missed signal:

- You conclude delivery times are fine, but they actually have gotten worse
- Consequence: customer dissatisfaction goes unaddressed

> **Trade-off:** Reducing α (fewer false alarms) increases β (more missed signals). Larger samples reduce both.

Step 3: Identify the Test Statistic

Test statistic: a value computed from sample data to decide whether to reject H_0.

For one-sample tests of a population mean:

z-test (σ known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

t-test (σ unknown)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

For a one-sample test about population variance, use the χ^2 statistic. Use the F-statistic when comparing two variances (or in ANOVA settings).

Step 4: Decision Rule

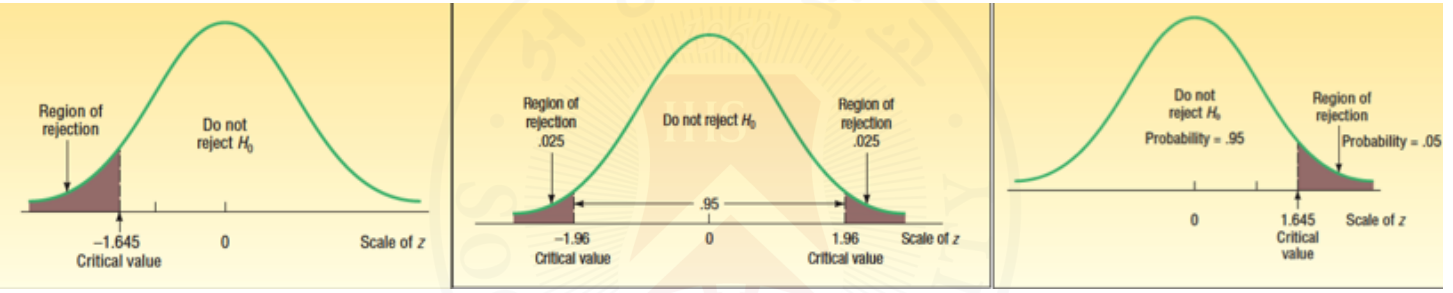
Two approaches:

1. **Critical value**
2. **p-value (preferred)**

1. Critical value

The **critical value** is the cutoff (or cutoffs) that separates:

- the **rejection region** (reject H_0), and
- the **non-rejection region** (do not reject H_0), for a chosen significance level α.



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Quick example (two-sided z-test):

- If $\alpha = 0.05$, critical values are $z = \pm 1.96$.
- If observed $z = 2.10$, then $2.10 > 1.96$.
- So the statistic is in the rejection region, and we **reject** H_0.

2. p-value (Important)

Definition:

The **p-value** is the probability of getting a test statistic as extreme as (or more extreme than) the observed one, **if H_0 is true**.

Quick example:

- Suppose a z-test gives $z = 2.00$.
- Right-tailed p-value: $P(Z \geq 2.00) \approx 0.0228$.
- Two-sided p-value: $2 \times 0.0228 = 0.0456$.

So, under H_0, a result this extreme would occur about **4.56%** of the time (two-sided case). Which means if we set $\alpha = 0.05$, we would reject H_0 because $p < \alpha$.

Decision Rule (p-value)

- If p-value < α → reject H0
- If p-value ≥ α → do not reject H0

Typical α:

- 0.05
- 0.01

Plain-English Guide to p-values

Interpretation (smaller p-value means stronger evidence against H_0):

- $p \leq 0.10$: weak to moderate evidence against H_0
- $p \leq 0.05$: moderate to strong evidence against H_0
- $p \leq 0.01$: strong evidence against H_0
- $p \leq 0.001$: very strong evidence against H_0

Decision reminder:

- Compare p-value with your pre-chosen α.
- If $p < \alpha$, reject H_0; otherwise, fail to reject H_0.

Step 5: Take a Sample, Arrive at a Decision

Checklist:

- Select a representative sample.
- Collect data on the relevant variable(s).
- Compute the test statistic (z or t).
- Compare with the critical value(s), or compare p-value with α.
- State the decision clearly: **reject H_0** or **fail to reject H_0**.

Decision template:

At significance level α, [reject / fail to reject] H_0.

Step 6: Interpret the Result

Translate the statistical decision into a plain-language business conclusion.

Checklist:

- State the decision first: **reject H_0** or **fail to reject H_0**.
- Refer to the original research question.
- Avoid saying H_0 is true; say "insufficient evidence" when failing to reject.
- If relevant, mention the significance level α.

Templates:

- Fail to reject: "At $\alpha = 0.05$, we fail to reject H_0; there is insufficient evidence that [effect/relationship] exists."
- Reject: "At $\alpha = 0.05$, we reject H_0; the data provide evidence that [effect/relationship] exists."

Example

A firm claims:

μ = 100 USD (average order)

Sample:

- $\bar{x}$ = 108, mean order value in the sample
- σ = 20, population standard deviation
- n = 25, sample size

Step 1: Define Hypotheses

H_0: $\mu = 100$ (the average order value is 100 USD)

H_1: $\mu \neq 100$ (the average order value is not 100 USD)

Step 2: Choose α and Test Type

Use a two-sided **z-test** because population σ is known and two-sides because the alternative hypothesis is two-tailed.

Set significance level: $\alpha = 0.05$

Step 3: Compute Test Statistic (z)

```
import math

xbar = 108
mu = 100
sigma = 20
n = 25

se = sigma / math.sqrt(n)
z = (xbar - mu) / se

z
```

```
## 2.0
```

$$z = \frac{108 - 100}{20/\sqrt{25}} = \frac{8}{4} = 2.00$$

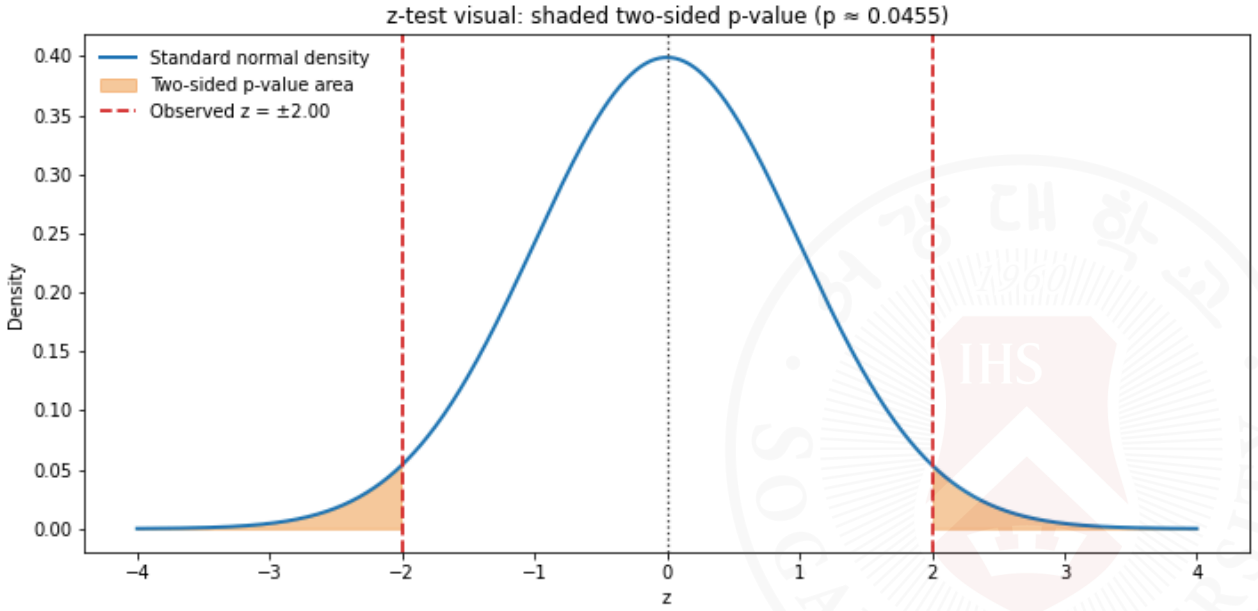
Step 4: Compute p-value

Observed statistic: $z = 2$

One-tail area: $P(Z > 2) \approx 0.0228$

Two-sided p-value: $p = 2 \times 0.0228 \approx 0.0456$

Visual Interpretation (z-test)



Step 5: Make the Decision

At α = 0.05:

p-value $= 0.0456 < 0.05 \rightarrow$ **reject** H_0

Step 6: Interpret in Business Terms

Conclusion:

At $\alpha = 0.05$, the data provide evidence that the average order value is **different from 100 USD**.

t-test Example (Same 6-Step Logic)

When σ is unknown, use a **t-test**.

Sample:

- x̄ = 42, sample mean
- s = 8, sample standard deviation
- n = 16, sample size

Step 1: $H_0 : \mu = 40$, $H_1 : \mu \neq 40$

Step 2: choose $\alpha = 0.05$ (two-sided)

```python
from scipy.stats import t
import math

xbar = 42
mu = 40
s = 8
n = 16

se = s / math.sqrt(n)
t_stat = (xbar - mu) / se

t_stat
```

```
## 1.0
```

$$t = \frac{42 - 40}{8/\sqrt{16}} = \frac{2}{2} = 1.00$$

Step 4 (t): Compute p-value

```
df = n - 1
p_value = 2 * (1 - t.cdf(abs(t_stat), df))

p_value
```

```
## np.float64(0.3331701359154764)
```

Interpretation

Step 5 (Decision):

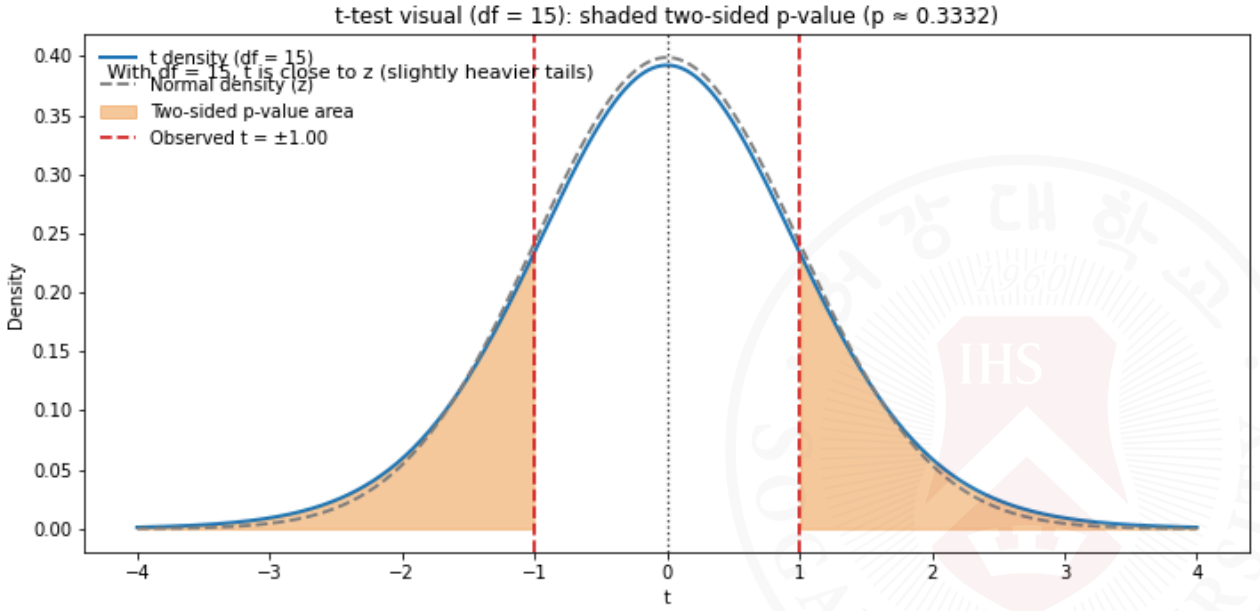
- If $p < 0.05$, reject H_0; otherwise, fail to reject H_0.

For this sample, $t = 1.00$ and the two-sided p-value is about 0.33, so we **fail to reject H_0** at $\alpha = 0.05$.

Step 6 (Interpretation):

- There is insufficient evidence that the mean differs from 40.

Visual Interpretation



Example: Jamestown Steel

Jamestown Steel changed its production methods and hired new employees.

Question: has the **mean weekly output** changed from 200 desks?

Given:

- population mean under H_0: $\mu_0 = 200$
- known population standard deviation: $\sigma = 16$
- sample size: $n = 50$ weeks
- sample mean: $\bar{x} = 203.5$
- significance level: $\alpha = 0.01$

Jamestown Steel: Step 1 and Step 2

Step 1: define the hypotheses

- $H_0 : \mu = 200$
- $H_1 : \mu \neq 200$

Because the question asks whether output **has changed**, this is a **two-sided** test.

Step 2: choose the significance level

- $\alpha = 0.01$

Jamestown Steel: Step 3

Use a **z-test** because the population standard deviation σ is known.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$z = \frac{203.5 - 200}{16/\sqrt{50}} \approx 1.55$$

Jamestown Steel: Steps 4, 5, and 6

Step 4: decision rule

- For a two-sided test with $\alpha = 0.01$, reject H_0 if $|z| > 2.576$.

Step 5: decision

- Observed $z = 1.55$
- Since $1.55 < 2.576$, we **fail to reject H_0**.

Step 6: interpretation

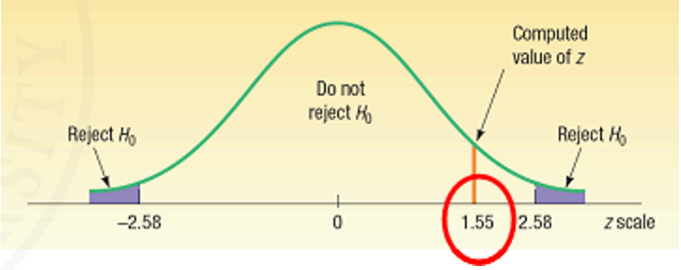
- At the 1% significance level, there is **insufficient evidence** that mean weekly production has changed from 200 desks.

$$|z| > z_{\alpha/2}$$

$$\left|\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}\right| > z_{\alpha/2}$$

$$\left|\frac{203.5-200}{16/\sqrt{50}}\right| > z_{.01/2}$$

1.55 is not > 2.58



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

One-Tailed Variation

Now suppose the VP asks a narrower question:

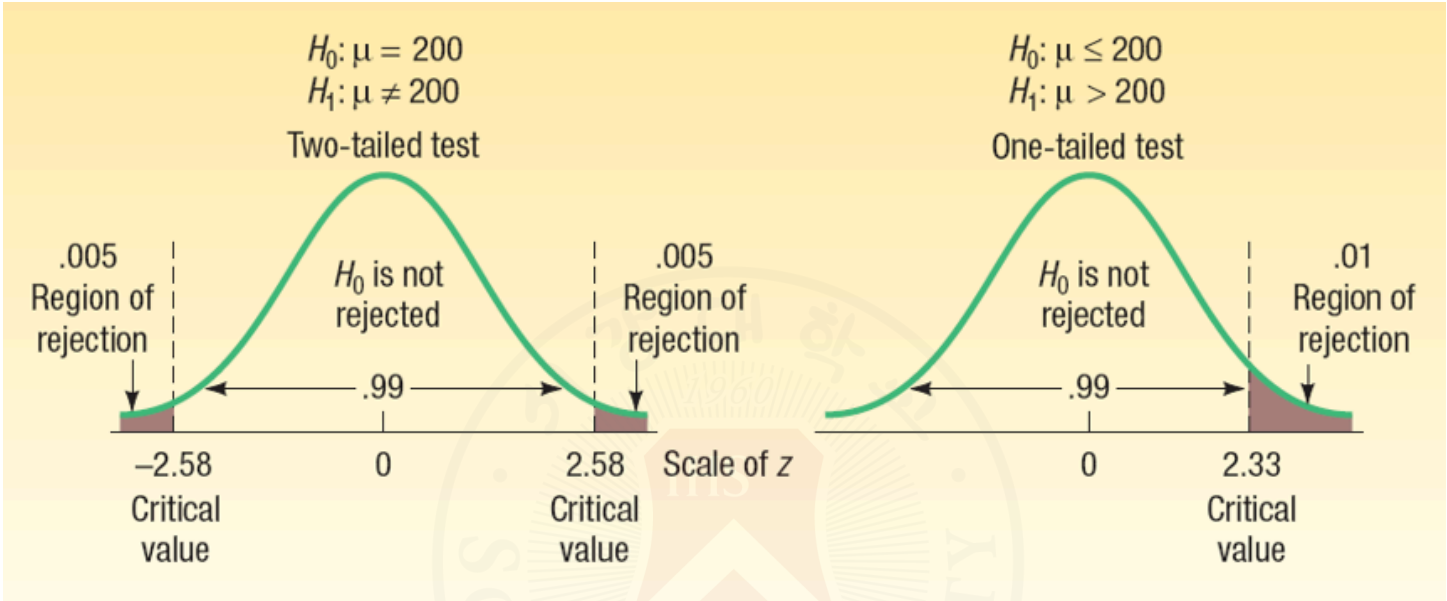
> Has production **increased** after the new methods were introduced?

Then the hypotheses become:

- $H_0 : \mu \leq 200$
- $H_1 : \mu > 200$

This is a **right-tailed** test.

One-Tailed Test versus Two-Tailed Test



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Key differences between one-tailed and two-tailed tests (same α):

- **Critical value:** In a two-tailed test, α is split equally across both tails ($\alpha/2$ each), so the cutoff is more extreme: $z = \pm 2.576$ at $\alpha = 0.01$. In a one-tailed test, the entire α is in one tail, giving a less extreme cutoff: $z = 2.33$ (right-tailed, $\alpha = 0.01$). It is therefore *easier* to reject H_0 with a one-tailed test.

One-Tailed Jamestown Test

Steps 1-3:

- $H_0 : \mu \leq 200$
- $H_1 : \mu > 200$
- $\alpha = 0.01$
- Use a z-test because σ is known.

Step 4: decision rule

- Reject H_0 if $z > 2.33$.

Step 5: decision

- Observed $z = 1.55$
- Since $1.55 < 2.33$, we **fail to reject H_0**.

Step 6: interpretation

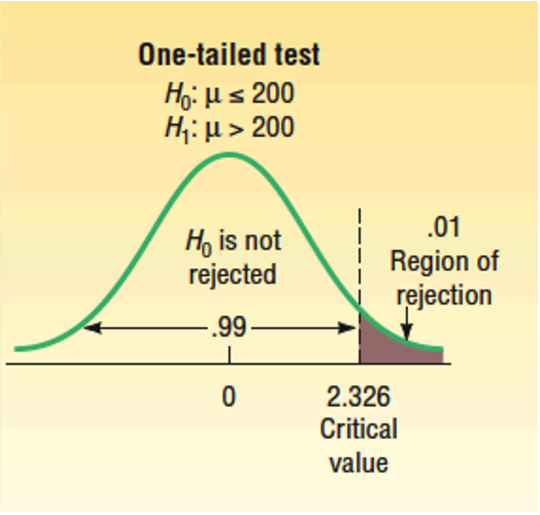
- Based on this sample, we cannot conclude that average weekly production increased.

$$z > z_\alpha$$

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} > z_\alpha$$

$$\frac{203.5 - 200}{16/\sqrt{50}} > z_{.01}$$

$$1.547 \text{ is not } > 2.326$$



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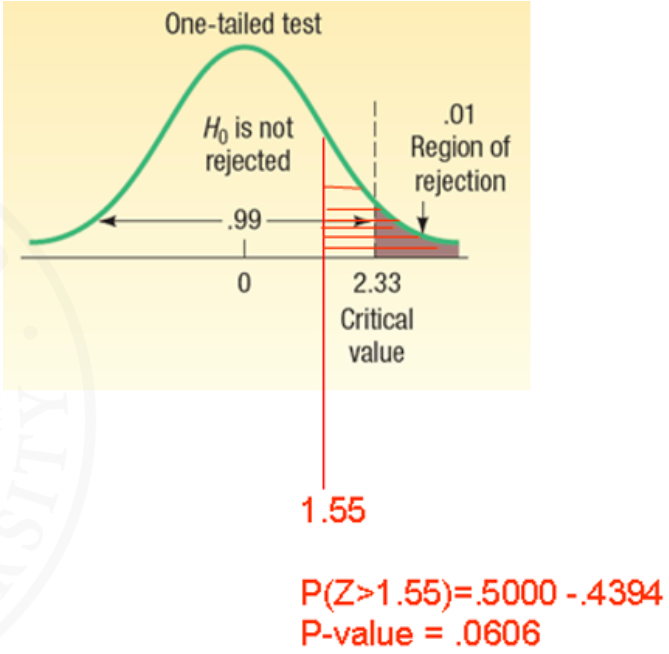
p-Value for the One-Tailed Test

For the right-tailed test:

- $H_0 : \mu \leq 200$
- $H_1 : \mu > 200$
- observed $z = 1.55$
- critical value at $\alpha = 0.01$: $z_\alpha = 2.33$

p-value approach:

- p-value ≈ 0.0606
- compare with $\alpha = 0.01$
- since $0.0606 > 0.01$, we **fail to reject H_0**



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Left-Tailed Example

Delivery-time claim: average delivery is **under 5 days**.

Given:

- $\mu_0 = 5$, the value under H_0
- $\sigma = 1.2$, population standard deviation
- $n = 36$, sample size
- $\bar{x} = 4.6$, sample mean
- $\alpha = 0.05$, significance level

Left-Tailed Example: Steps 1-3

Step 1 (hypotheses):

- $H_0 : \mu \geq 5$
- $H_1 : \mu < 5$

Step 2 (significance level):

- $\alpha = 0.05$

Step 3 (test statistic, z-test since σ known):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{4.6 - 5}{1.2/\sqrt{36}} = -2.00$$

Left-Tailed Example: Steps 4-6

Step 4 (decision rule):

- For a left-tailed test at $\alpha = 0.05$, reject H_0 if $z < -1.645$.

Step 5 (decision):

- Observed $z = -2.00 < -1.645 \rightarrow$ **reject H_0**.
- Equivalent p-value is about $0.0228 < 0.05$.

Step 6 (interpretation):

- At the 5% level, the data provide evidence that mean delivery time is **below 5 days**.

Key Differences: z vs t

Feature	z	t
σ known	Yes	No
Sample size	Any	Small OK
Shape	Fixed	Depends on df

Common Mistakes

- confusing CI with hypothesis testing
- wrong test (z vs t)
- incorrect interpretation of p-value

In-Class Practice

A company claims:

μ = 50

Sample:

- x̄ = 54
- s = 12
- n = 36

Tasks:

1. Step 1: State H_0 and H_1.
2. Step 2: Set $\alpha = 0.05$ and choose z or t.
3. Step 3: Compute the test statistic.
4. Step 4: Use p-value or critical value decision rule.
5. Step 5: State the decision (reject/fail to reject H_0).
6. Step 6: Write one business interpretation sentence.

Final Idea

Hypothesis testing helps us:

move from data → decision

Summary

- H0 vs H1: competing claims about a population parameter
- test statistic: z (σ known) or t (σ unknown)
- p-value: probability of observing data as extreme as ours if H0 is true
- decision rule: reject H0 if $p < \alpha$; otherwise, fail to reject H0
- interpretation: "At $\alpha = 0.05$, there is [sufficient/insufficient] evidence to conclude that [effect/relationship] exists."

Next Week

(May 5) Two-sample hypothesis testing (LMW Chapter 11) - **Public Holiday** - no class - makeup session with recorded lecture and assignment (Cyber Campus)

(May 7) Nominal-level hypothesis tests (LMW Chapter 15)

Any questions?

Thank you for your attention and active participation!